



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Advanced Natural Language Processing

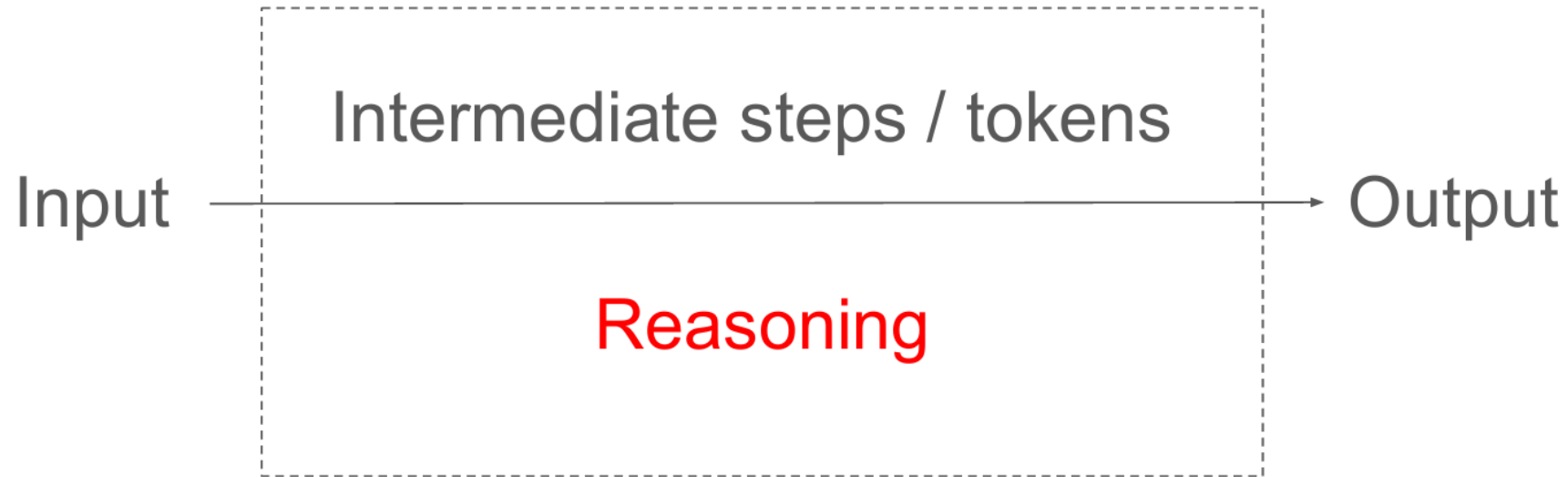
Lecture 21: Reasoning LLMs



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What is LLM Reasoning?



Ling et al. Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems. ACL 2017
Using intermediate steps or program traces was widely adopted in neural-symbolic literature, e.g., neural-symbolic stack machines (NeurIPS 2020)
by Compositional generalization via Chen et al.
Stanford, CS25: Transformers United V5, April 29, 2025

What is LLM Reasoning?



What is the output when concatenating the last letter of each word in “artificial intelligence”?

No reasoning

The answer is “le”.

Reasoning



The last letter of “artificial” is “l”. The last letter of “intelligence” is “e”. Concatenating “l” and “e” leads to “le”. So the answer is “le”.

What is LLM Reasoning?

Problem 1:

Question: Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

Options: A) $3/7$ B) $3/2$ C) $3/88$ D) $3/8$ E) $2/2$

Rationale: Let the speeds of the two trains be x m/sec and y m/sec respectively. Then, length of the first train = $27x$ meters, and length of the second train = $17y$ meters. $(27x + 17y) / (x + y) = 23 \rightarrow 27x + 17y = 23x + 23y \rightarrow 4x = 6y \rightarrow x/y = 3/2$.

Correct Option: B

What is LLM Reasoning?

GSM8K: <Problem, **Intermediate Steps**, Answer>

Following the work by Ling et al 2017, Cobbe et al 2021 in OpenAI built a much larger math word problem dataset (GSM8K) with natural language rationales, and used it to finetune GPT3

Problem: Ali is a dean of a private school where he teaches one class. John is also a dean of a public school. John has two classes in his school. Each class has $\frac{1}{8}$ the capacity of Ali's class which has the capacity of 120 students. What is the combined capacity of both schools?

Solution: Ali's class has a capacity of 120 students. Each of John's classes has a capacity of $120/8 = 15$ students. The total capacity of John's two classes is $15 \text{ students} * 2 \text{ classes} = 30 \text{ students}$. The combined capacity of the two schools is $120 \text{ students} + 30 \text{ students} = 150 \text{ students}$.

Final answer: 150



Cobbe et al. Training Verifiers to Solve Math Word Problems. [arXiv:2110.14168](https://arxiv.org/abs/2110.14168) [cs.LG]. 2021

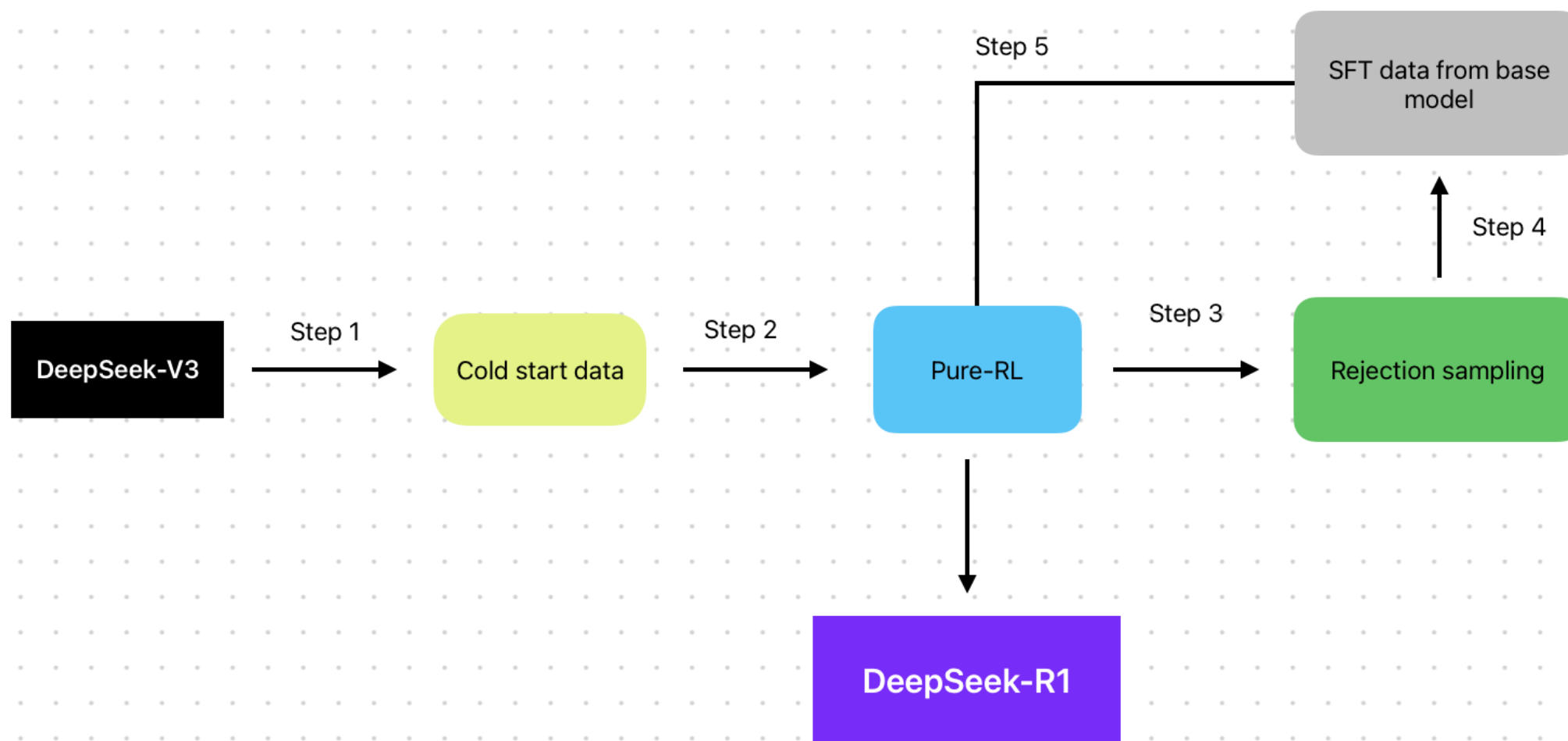


DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

research@deepseek.com

Training DeepSeek-R1



Group Relative Policy Optimization



- GRPO algorithm is based on the Proximal Policy Optimization (PPO) algorithm.
- Reward model is a trained model that assigns rewards.
- Value model estimates the long-term expected future reward from a given state.

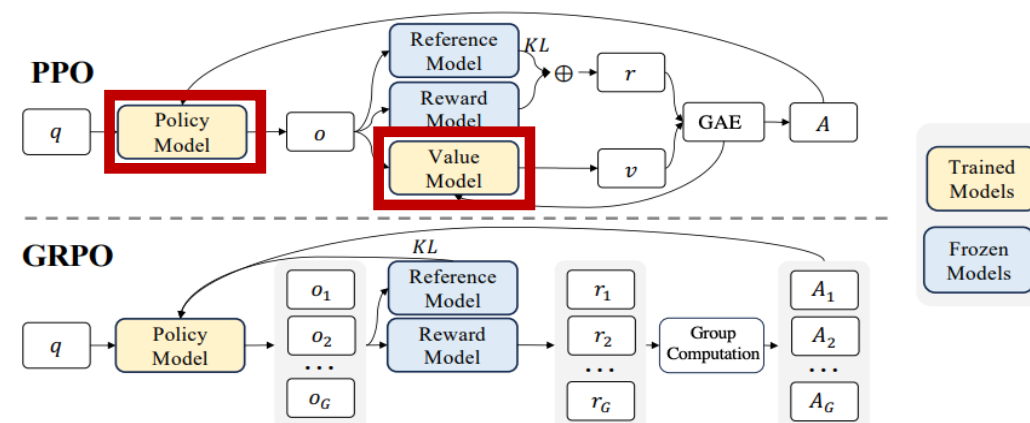


Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.

Group Relative Policy Optimization



- GRPO generates multiple responses at once and generates multiple rewards.
- Group computation decides the advantage of the response based on other responses in the group.
- The value model is removed since it's expensive to evaluate multiple actions every state.

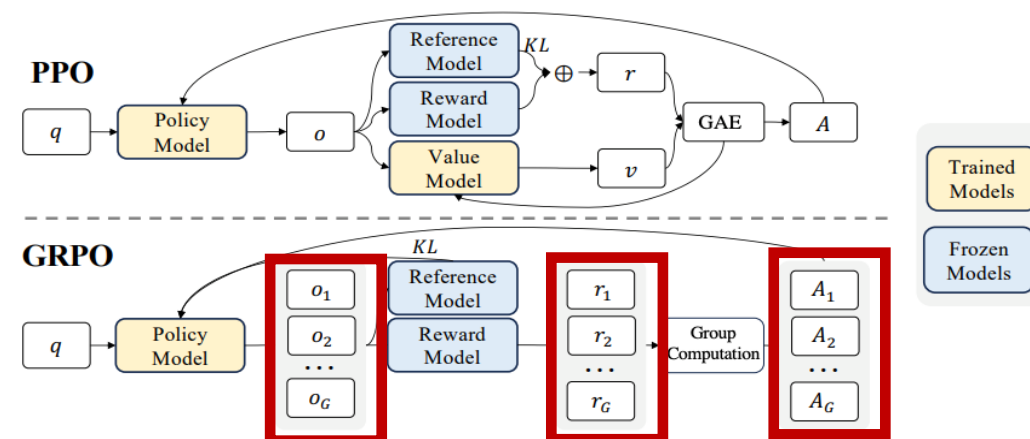


Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.

Group Relative Policy Optimization



- Optimizes the policy to maximize the objective J_{GRPO} .
- For every question (q), the model generates multiple outputs (o_i).
- Scale the reward by the ratio or clipped ratio (conservative updates).
- Maintain divergence from the ref model (base model).

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}.$$

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}_{[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]} \left(\frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \varepsilon, 1 + \varepsilon \right) A_i \right) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right) \right), \quad (1)$$

Reward Modeling in GRPO



- Uses a rule-based reward system with two types:
 - Accuracy rewards: Ensures correct responses (e.g., math, LeetCode).
 - Format rewards: Employ a format reward model that enforces the model to put its thinking process between '`<think>`' and '`</think>`' tags.

DeepSeek-R1-Zero Response Length

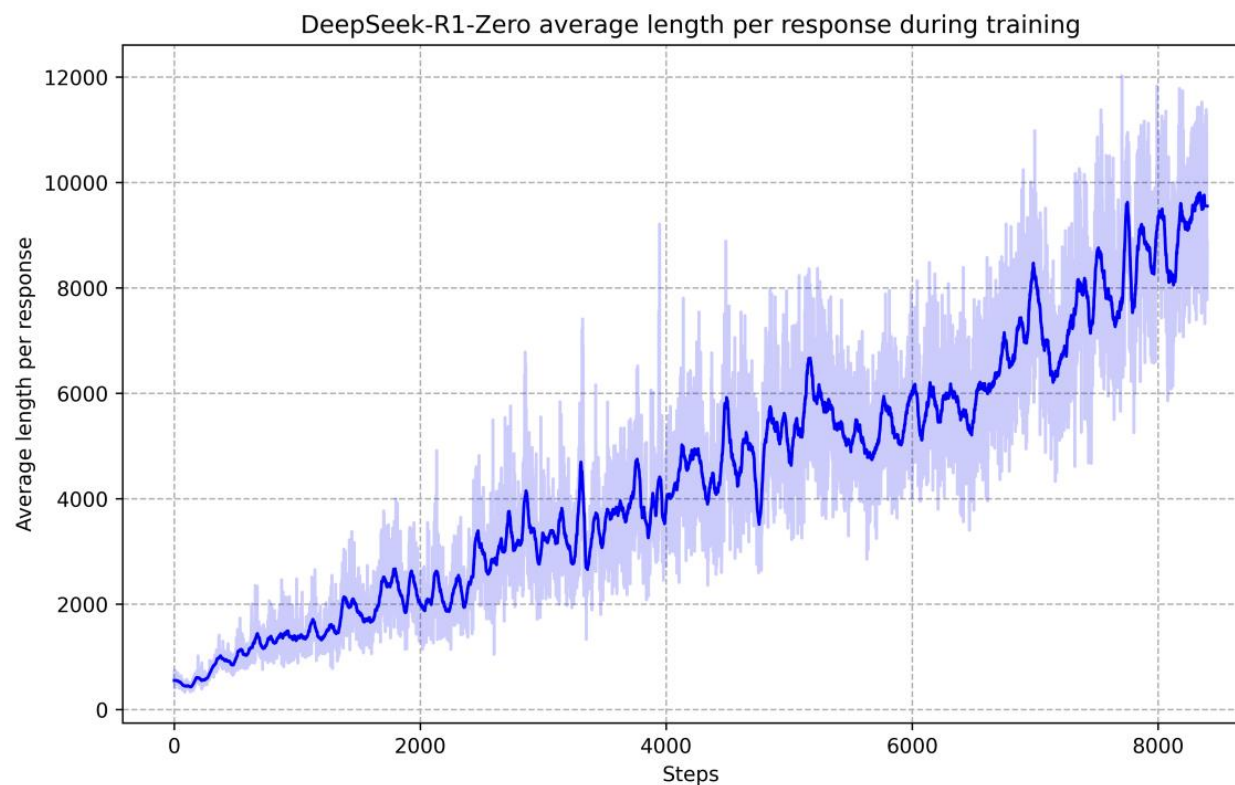


Figure 3 | The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time.

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444

Table 2 | Comparison of DeepSeek-R1-Zero and OpenAI o1 models on reasoning-related benchmarks.

Aha Moment of DeepSeek-R1-Zero



Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

Table 3 | An interesting “aha moment” of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

DeepSeek-R1 Achieves SOTA Performance



Benchmark (Metric)		Claude-3.5- Sonnet-1022	GPT-4o 0513	DeepSeek V3	OpenAI o1-mini	OpenAI o1-1217	DeepSeek R1
English	Architecture	-	-	MoE	-	-	MoE
	# Activated Params	-	-	37B	-	-	37B
	# Total Params	-	-	671B	-	-	671B
	MMLU (Pass@1)	88.3	87.2	88.5	85.2	91.8	90.8
	MMLU-Redux (EM)	88.9	88.0	89.1	86.7	-	92.9
	MMLU-Pro (EM)	78.0	72.6	75.9	80.3	-	84.0
	DROP (3-shot F1)	88.3	83.7	91.6	83.9	90.2	92.2
	IF-Eval (Prompt Strict)	86.5	84.3	86.1	84.8	-	83.3
	GPQA Diamond (Pass@1)	65.0	49.9	59.1	60.0	75.7	71.5
	SimpleQA (Correct)	28.4	38.2	24.9	7.0	47.0	30.1
Code	FRAMES (Acc.)	72.5	80.5	73.3	76.9	-	82.5
	AlpacaEval2.0 (LC-winrate)	52.0	51.1	70.0	57.8	-	87.6
	ArenaHard (GPT-4-1106)	85.2	80.4	85.5	92.0	-	92.3
	LiveCodeBench (Pass@1-COT)	38.9	32.9	36.2	53.8	63.4	65.9
	Codeforces (Percentile)	20.3	23.6	58.7	93.4	96.6	96.3
Math	Codeforces (Rating)	717	759	1134	1820	2061	2029
	SWE Verified (Resolved)	50.8	38.8	42.0	41.6	48.9	49.2
	Aider-Polyglot (Acc.)	45.3	16.0	49.6	32.9	61.7	53.3
Chinese	AIME 2024 (Pass@1)	16.0	9.3	39.2	63.6	79.2	79.8
	MATH-500 (Pass@1)	78.3	74.6	90.2	90.0	96.4	97.3
	CNMO 2024 (Pass@1)	13.1	10.8	43.2	67.6	-	78.8
Chinese	CLUEWSC (EM)	85.4	87.9	90.9	89.9	-	92.8
	C-Eval (EM)	76.7	76.0	86.5	68.9	-	91.8
	C-SimpleQA (Correct)	55.4	58.7	68.0	40.3	-	63.7

Table 4 | Comparison between DeepSeek-R1 and other representative models.

Distilled Model Evaluation

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

Table 5 | Comparison of DeepSeek-R1 distilled models and other comparable models on reasoning-related benchmarks.

Takeaways



- DeepSeek-R1 matches OpenAI o1 in complex reasoning benchmarks
- DeepSeek-R1-zero achieves strong logical reasoning without conventional supervised fine-tuning on reasoning tasks
- Distillation enables smaller models to inherit advanced reasoning capabilities from DeepSeek-R1

Further Reading



- [What is Reinforcement Learning? — My sample book](#)

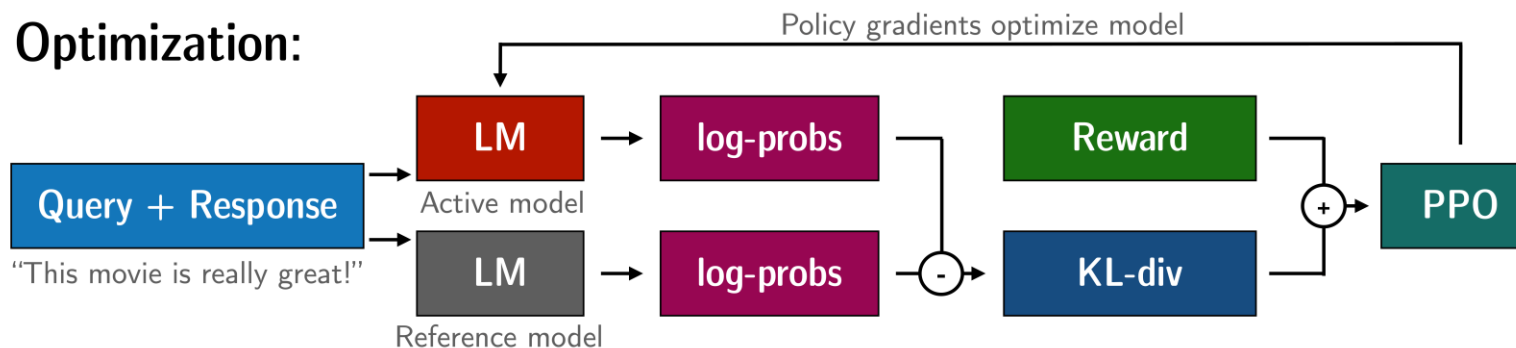
Rollout:



Evaluation:



Optimization:



Further Reading

- [GRPO, simplified](#), [Implementing GRPO](#)
- [Introduction - Hugging Face Deep RL Course](#)
- [BytedTsinghua-SIA/DAPPO: An Open-source RL System from ByteDance Seed and Tsinghua AIR](#)

CoT Reasoning Without Prompting

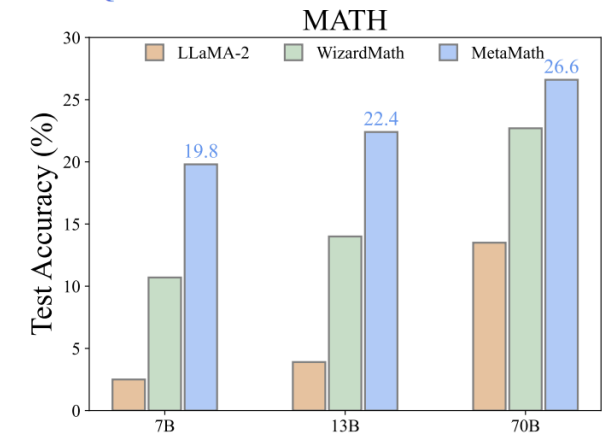
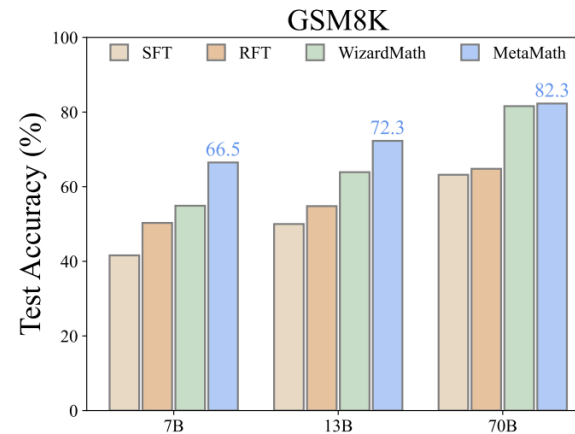
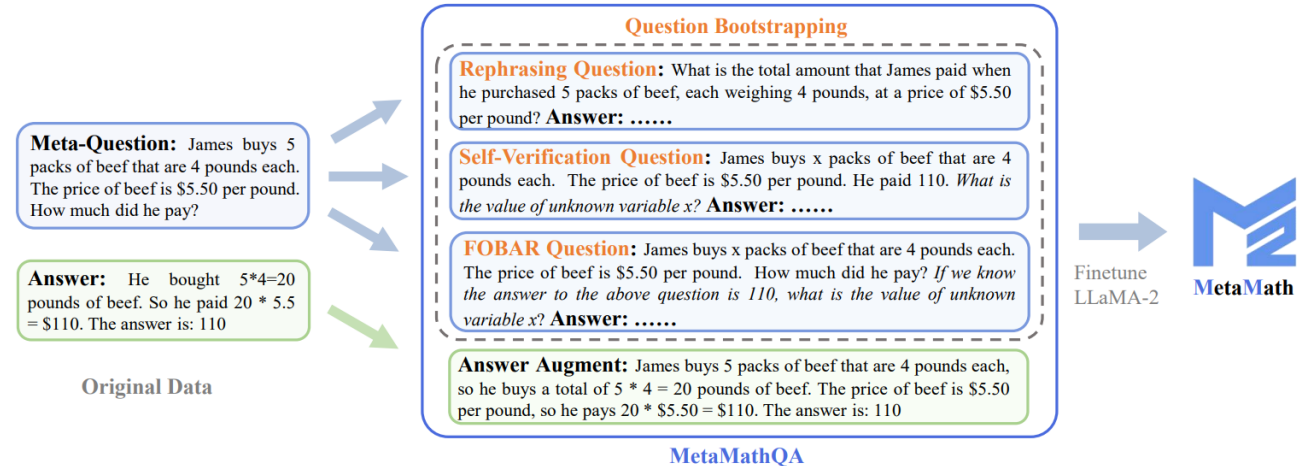
Chain-of-Thought Decoding

-
1. Go beyond greedy decoding by checking more generation candidates
 2. Choose candidates which have the highest confidence on the final answer
-

- How can we train a model to learn cot reason?
 - 1. High-quality reasoning data
 - 2. Novelty Training Methods

High-quality reasoning data

- MetaMathQA expands beyond traditional answer augmentation to generate diverse question formulations that capture the same mathematical concepts from multiple perspectives



MetaMath: Bootstrap Your Own Mathematical Questions for Large Language Models (ICLR 2024 Spotlight)

High-quality reasoning data

AI-MO/NuminaMath-CoT

- Approximately 860k math problems, where each solution is formatted in a Chain of Thought (CoT) manner.
- The processing steps include
 - (a) OCR from the original PDFs
 - (b) segmentation into problem-solution pairs
 - (c) Translation into English
 - (d) Realignment to produce a CoT reasoning format, and
 - (e) final answer formatting

Datasets: AI-MO/NuminaMath-CoT  like 507  Follow Project-Numina 434

Tasks:  Text Generation Modalities:  Text Formats:  parquet Languages:  English Size: 100K - 1M Tags: aimo math Libraries

[Dataset card](#) [Data Studio](#) [Files and versions](#)  xet  Community 8

Dataset Viewer Auto-converted to Parquet [API](#) [Embed](#) [Data Studio](#)

Split (2)
train · 859k rows

Search this dataset

source string · classes	problem string · lengths	solution string · lengths	messages list · lengths
9 values	10 6.1k	13 10.4k	2
cn_k12	Factorize: $x^3 - 6x^2 + 9x = \dots$	To factorize the given expression $x^3 - 6x^2 + 9x$, we start by identifying a common factor...	[{ "content": "F... $x^3 - 6x^2 + 9x =$
olympiads	In triangle ABC , $CA = CB$, and D is the midpoint of AB . Line EF passes...	1. Identify Key Elements : Consider $\triangle ABC$ where $CA = CB$ and D is the...	[{ "content": "I... and D is the mi
synthetic_amc	Consider the product of x^5 , $x^2 + \frac{1}{x^2}$, and $1 + \frac{1}{x}$ +...	1. Multiply the first two expressions : $(x^5 \left(x^2 + \frac{1}{x^2} \right)) = x^5 \cdot x^2 + \dots$	[{ "content": "C... $x^2 + \frac{1}{x^2}$
synthetic_math	The sum of three numbers a , b , and c is 120. If we increase a by 8, decrease...	We start with the equations: 1. $(a + b + c = 120)$ 2. $(a + 8 + b - 3 = 3c)$ First, solve $(b - 3 = \dots$	[{ "content": "T... b , and c is 1
orca_math	A vessel of capacity 2 liters has a certain percentage of alcohol and another...	Let's denote the percentage of alcohol in the first vessel (2 liters) as $x\%$. From the second vessel (6...	[{ "content": "A... has a certain per
	Given that the sum of the first n terms	Since $S_n = 2^n - 1$, we have $S_{n-1} = 2^{n-1} - 1$	[{ "content": "G...

< Previous 1 2 3 ... 8,595 Next >

R1-Distill Dataset

zwhe99/DeepMath-103K

- DeepMath-103K is meticulously curated to push the boundaries of mathematical reasoning in language models. Key features include:

- 1. Challenging Problems
- 2. Data Diversity and Novelty

Datasets: zwhe99/DeepMath-103K like 274

Tasks: Text Generation Modalities: Text Formats: parquet Languages: English Size: 100K - 1M ArXiv: arxiv:2504.11456

Libraries: Datasets Dask Croissant +1 License: mit

Dataset card Data Studio Files and versions xet Community 4

Dataset Viewer Auto-converted to Parquet API Embed Data Studio

Split (1)
train · 103k rows

Search this dataset

question	final_answer	difficulty	topic	r1_solution_1	r1_solution_2
string	string	float64	string	string	string
$\lim_{x \rightarrow \infty} \frac{1}{x}$	∞	5	Mathematics	Okay, so I need to find the limit as x approaches...	Okay, so I need to find the limit as x approaches infinity of (x!) raised to the...
"Find the length of the polar curve...	$\frac{\pi}{2}$	5	Mathematics -> Calculus...	"Okay, so I need to find the length of the polar...	"Okay, so I need to find the length of the polar curve given by $r = \sqrt{1 - \cos(2\theta)}$...
"Let A be a proper infinite...	$ A $	3	Mathematics -> Algebra ...	"Okay, let's try to figure out the...	"Okay, so I have this problem here: Let A be a proper infinite subset of a set X ...
"Consider a regular pentagon...	ϕ^4	5.5	Mathematics -> Geometry...	"Okay, so I have this problem about a regular...	"Okay, so I have this problem about a regular pentagon circumscribed in a...
"Find the longest geometric...	6	4	Mathematics -> Algebra ...	"Okay, so I need to find the longest geometric...	"Okay, so I need to find the longest geometric progression (GP) with a common...
"Find the expectation $E[X_t]$...	0	8	Mathematics -> Applied...	"Okay, so I need to find the expectation of X_t ...	"Okay, so I need to find the expectation of $X_t = \sin(B_t)$, where B_t is a Brownian motion...

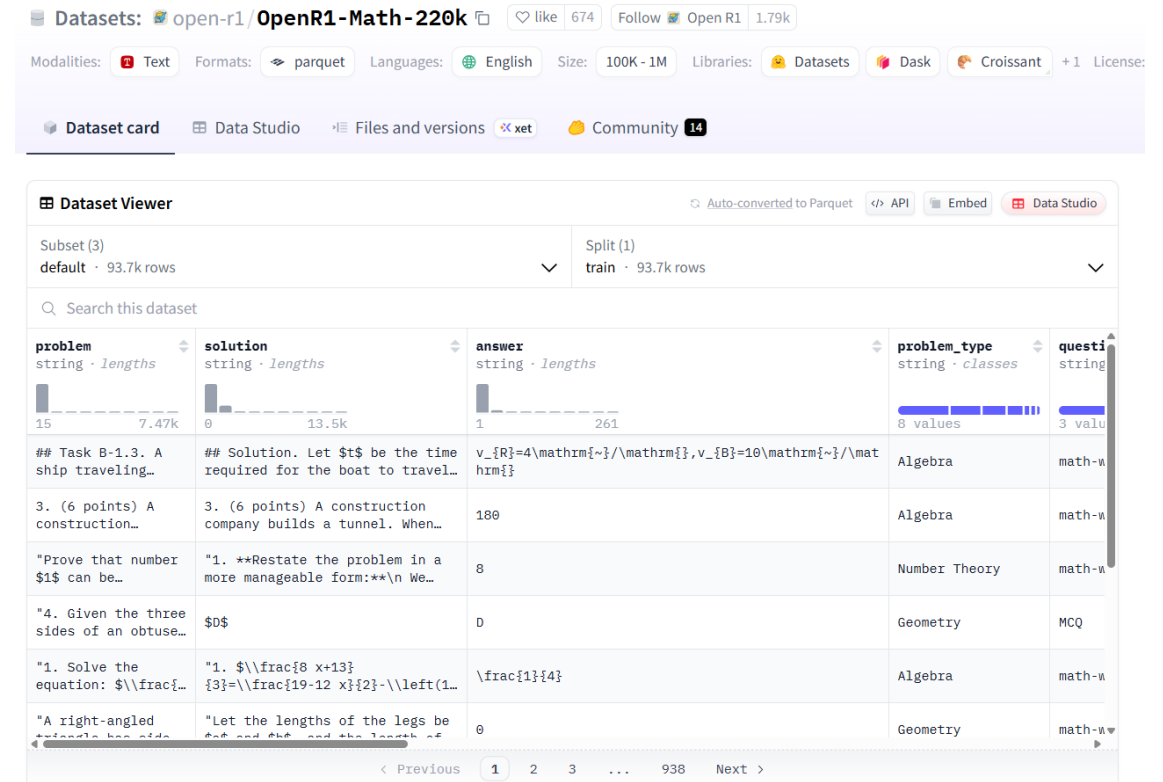
End of preview. Expand in Data Studio

< Previous 1 2 3 ... 1,031 Next >

R1-Distill Dataset

open-r1/OpenR1-Math-220k

- OpenR1-Math-220k is a large-scale dataset for mathematical reasoning.
- It consists of 220k math problems with two to four reasoning traces generated by [DeepSeek R1](#) for problems from NuminaMath 1.5.
- The traces were verified using [Math Verify](#) for most samples and [Llama-3.3-70B-Instruct](#) as a judge for 12% of the samples, and each problem contains at least one reasoning trace with a correct answer



The screenshot shows the Hugging Face Datasets page for 'open-r1/OpenR1-Math-220k'. It displays dataset statistics, modalities (Text), formats (parquet), languages (English), size (100K - 1M), and libraries (Datasets, Dask, Croissant). Below the header is the 'Dataset Viewer' showing a subset of 93.7k rows. The viewer includes a search bar and a table with columns: problem, solution, answer, problem_type, and question. The table shows several rows of math problems and their corresponding solutions and answers.

problem	solution	answer	problem_type	question
## Task B-1.3. A ship traveling...	## Solution. Let t be the time required for the boat to travel...	$v_{\{R\}}=4\mathrm{m}/\mathrm{s}, v_{\{B\}}=10\mathrm{m}/\mathrm{s}$	Algebra	math-w
3. (6 points) A construction...	3. (6 points) A construction company builds a tunnel. When...	180	Algebra	math-w
"Prove that number $\$1$ can be...	"1. **Restate the problem in a more manageable form:**\n We...	8	Number Theory	math-w
"4. Given the three sides of an obtuse...	$\$0$	0	Geometry	MCQ
"1. Solve the equation: $\frac{8x+13}{3}=\frac{19-12x}{2}-\left(1...$	"1. $\frac{8x+13}{3}=\frac{19-12x}{2}-\left(1...$	$\frac{1}{4}$	Algebra	math-w
"A right-angled triangle has side...	"Let the lengths of the legs be a and b , and the length of...	0	Geometry	math-w

- What problems exist with the current SFT?

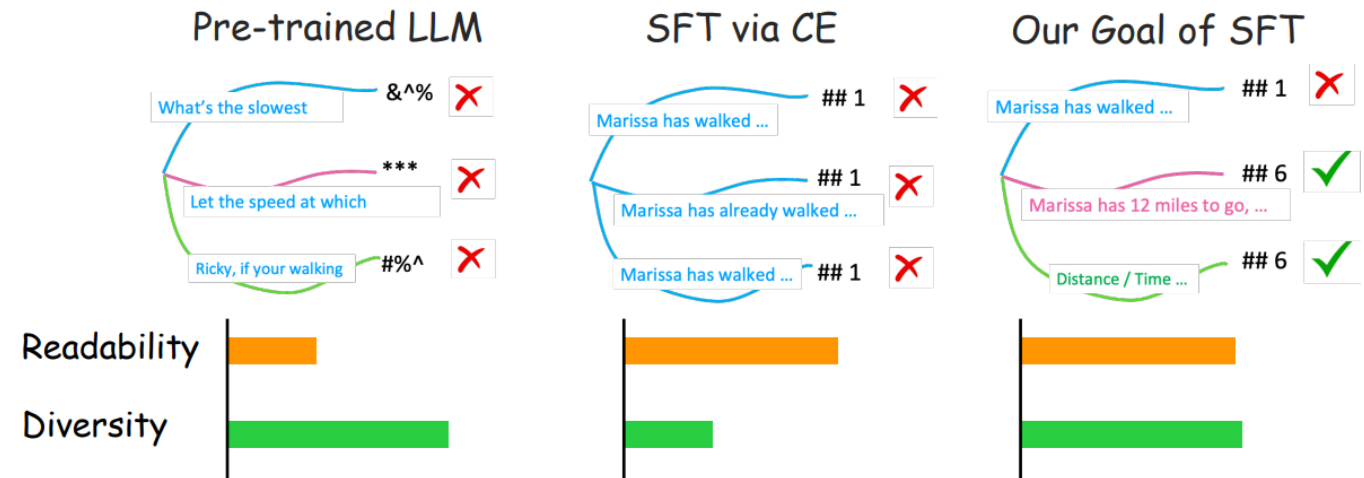
$$\min_{\theta} \mathcal{L}_{\text{CE}}(\theta) = -\mathbb{E}_{x \sim \rho} \mathbb{E}_{y \sim p(\cdot|x)} [\log f_{\theta}(y|x)].$$

- 1. SFT treats all tokens equally.
- 2. SFT reduces output diversity.

- Preserving Diversity in Supervised Fine-Tuning of Large Language Models. ICLR 2025.

Question: Marissa is hiking a 12-mile trail. She took 1 hour to walk the first 4 miles, then another hour to walk the next two miles. If she wants her average speed to be 4 miles per hour, what speed (in miles per hour) does she need to walk the remaining distance?

Answer: ## 6



[Preserving Diversity in Supervised Fine-Tuning of Large Language Models | OpenReview](#)

Figure 1: Illustration of diversity preservation in SFT. While pre-trained LLMs produce diverse outputs, these often lack proper formatting. Standard SFT using CE improves readability but reduces diversity. We aim to maintain output diversity while enhancing the readability of LLMs' responses.

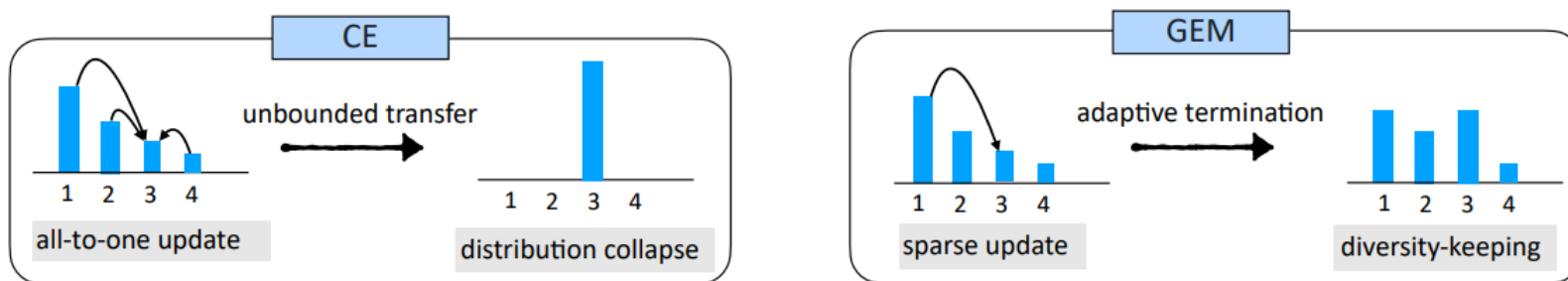
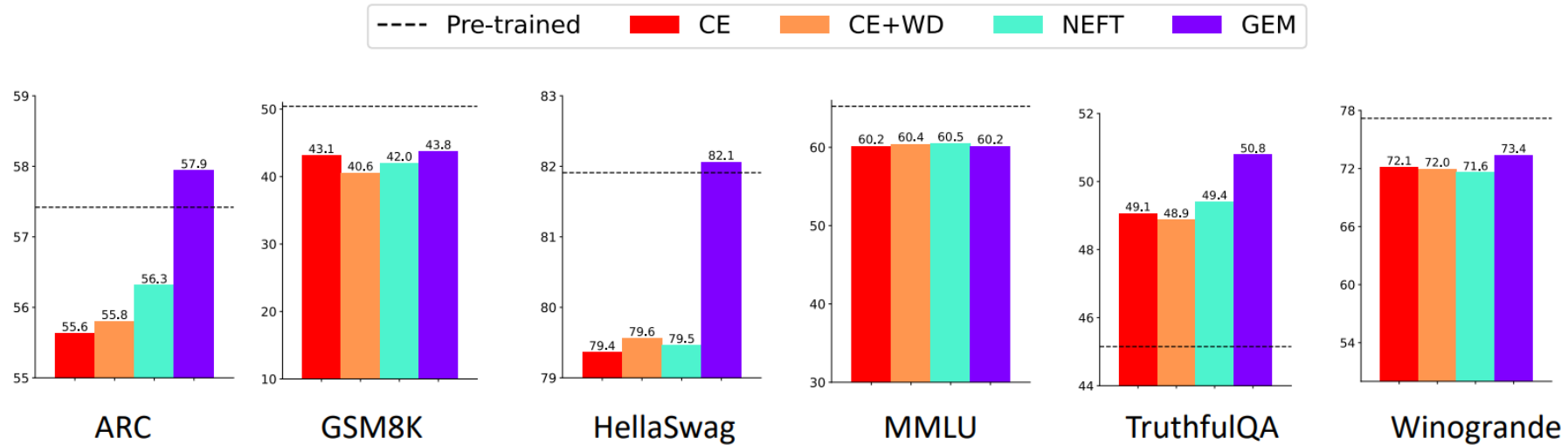


Figure 2: Comparison of learning schemes: CE v.s. GEM ($\beta = 0$). The arrows illustrate the probability movement directions during the learning process, with Token 3 as the target token.

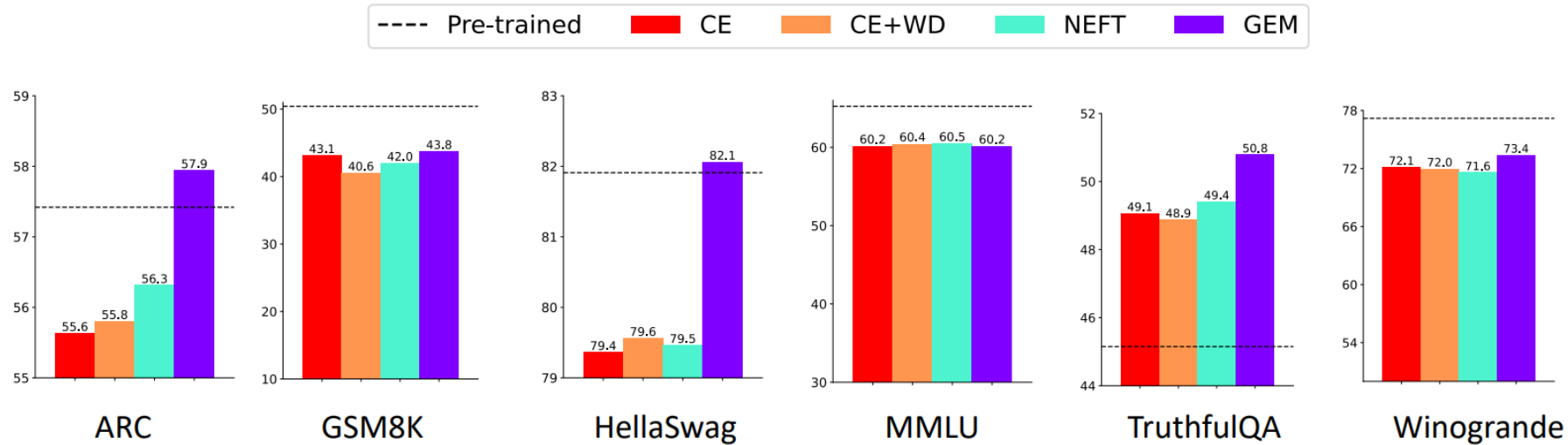
- (GEM)
- While the target token $i \notin \operatorname{argmax} f_{\theta_k}(\cdot|x)$, continue the following steps.
- Calculate the model's best prediction $j = \operatorname{argmax} f(\cdot|x)$
 - Decrease the logit for source token j by learning rate η and weight $w_{i \leftarrow j}$:

$$\theta_{k+1}[j] = \theta_k[j] - \eta * w_{i \leftarrow j}$$
 - Increase the logit for the target token i in a similar manner:

$$\theta_{k+1}[i] = \theta_k[i] + \eta * w_{i \leftarrow j}$$



Performance on tasks from the OpenLLM leaderboard. The results indicate that GEM outperforms CE, demonstrating a lower alignment tax.



Performance on tasks from the OpenLLM leaderboard. The results indicate that GEM outperforms CE, demonstrating a lower alignment tax.

- While SFT is widely adopted for its simplicity and efficiency in adapting models to expert demonstrations, it suffers from poor generalization to out-of-distribution examples and tends to overfit training data.

$$\mathcal{L}_{\text{SFT}}(\theta) = \mathbb{E}_{(x, y^*) \sim \mathcal{D}} [-\log \pi_{\theta}(y^* | x)].$$

Its gradient is:

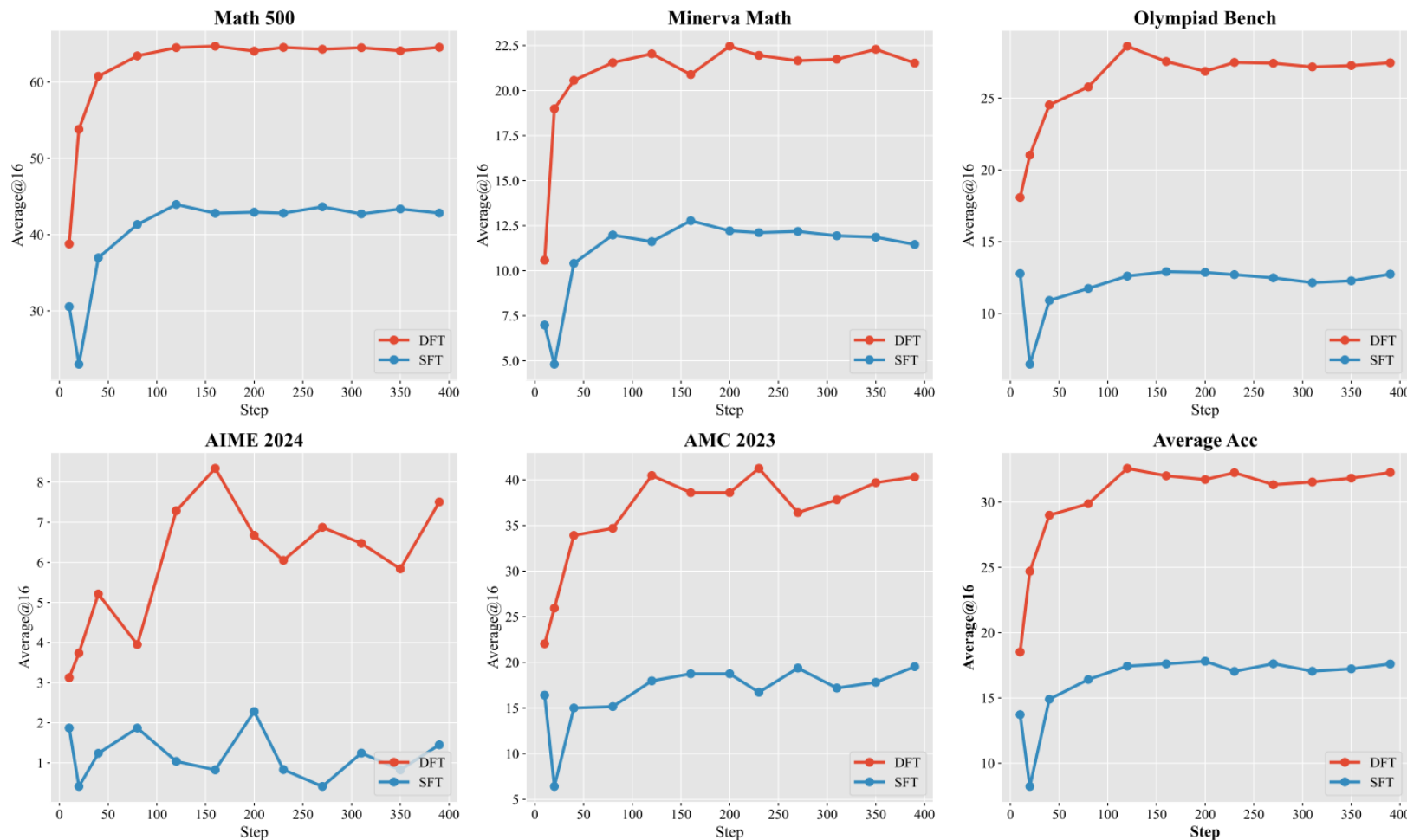
$$\nabla_{\theta} \mathcal{L}_{\text{SFT}}(\theta) = \mathbb{E}_{(x, y^*) \sim \mathcal{D}} [-\nabla_{\theta} \log \pi_{\theta}(y^* | x)].$$

[\[2508.05629\] On the Generalization of SFT: A Reinforcement Learning Perspective with Reward Rectification](#)

- The DFT proposes dynamically reweighting rewards by multiplying by a corrective inverse ratio (i.e., the policy probability itself). Specifically, the DFT modifies the gradient update of the SFT as follows:

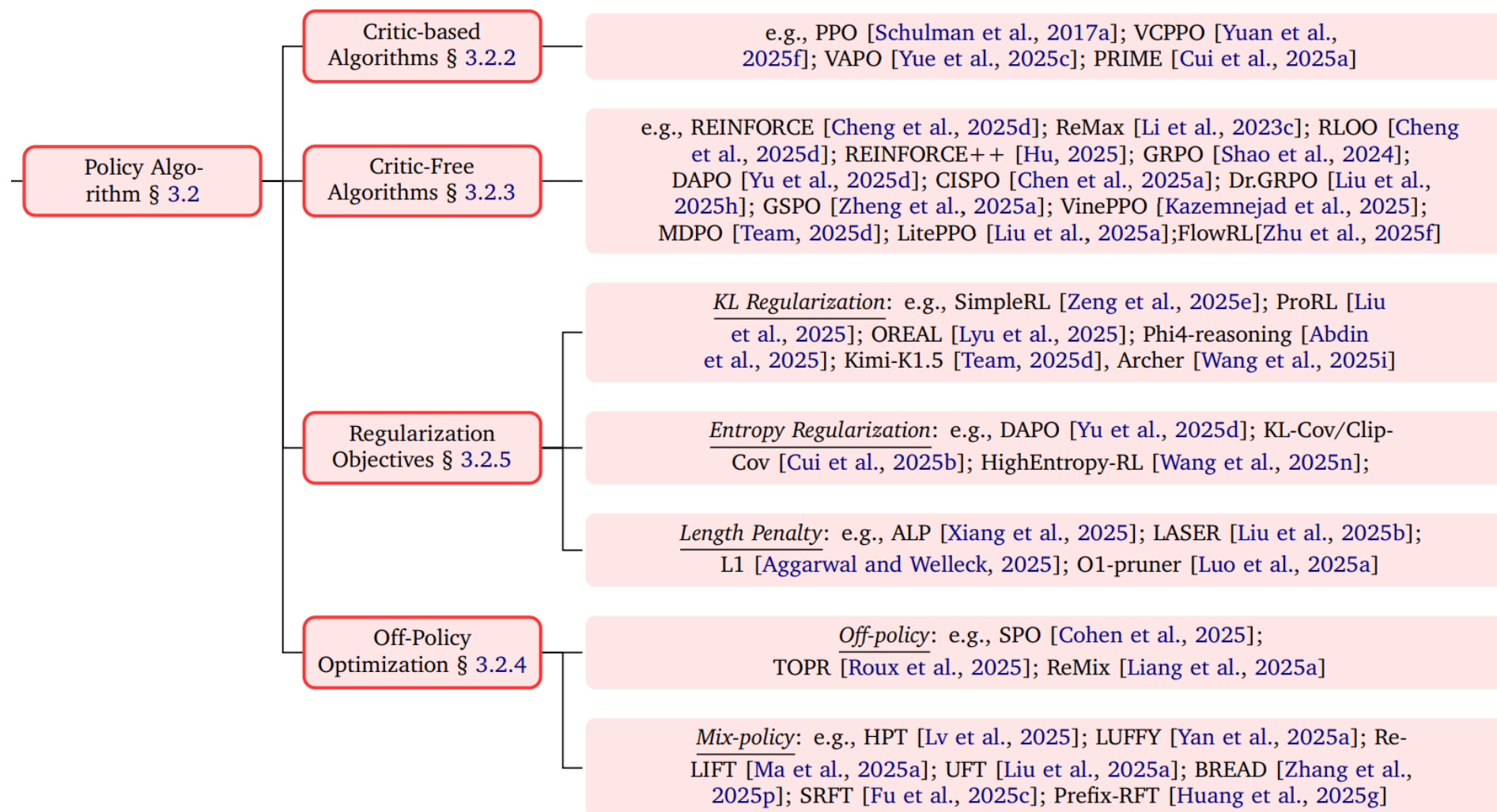
$$\mathcal{L}_{\text{DFT}}(\theta) = \mathbb{E}_{(x, y^*) \sim \mathcal{D}} \left[- \sum_{t=1}^{|y^*|} \text{sg} \left(\pi_{\theta}(y_t^* \mid y_{<t}^*, x) \right) \log \pi_{\theta}(y_t^* \mid y_{<t}^*, x) \right].$$

- This formula can be understood as follows: for each token, its loss contribution is weighted by its current predicted probability.



Accuracy progression for Qwen2.5-Math-1.5B across mathematical benchmarks, illustrating faster convergence and better performance achieved by DFT relative to SFT.

Reinforcement Learning



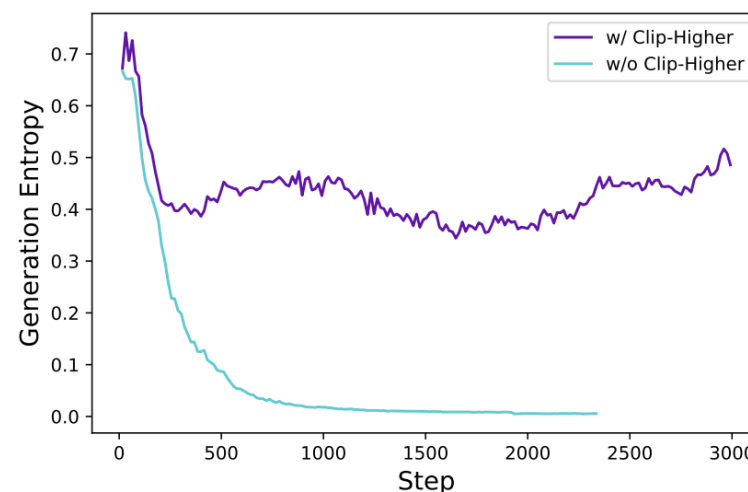
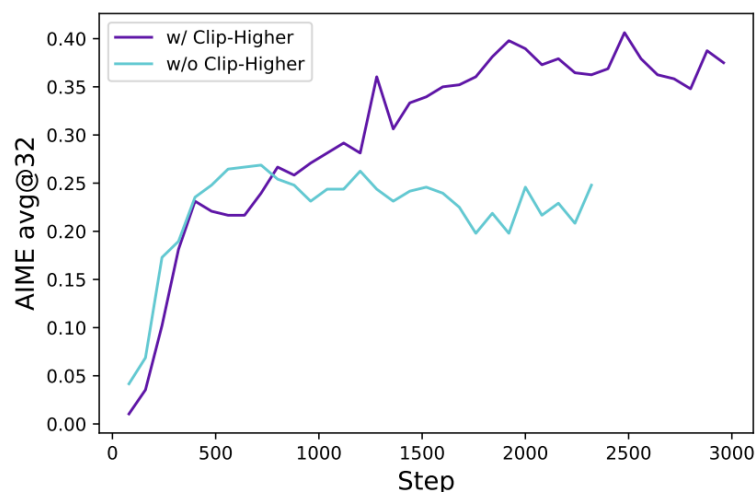
DAPO: An Open-Source LLM Reinforcement Learning System at Scale

- Challenges in Large-Scale RL for Reasoning Context
 - Entropy Collapse
 - Problem: Policy rapidly converges to high-probability tokens.
 - Impact: Becomes deterministic too early, preventing exploration of diverse reasoning paths.
 - Gradient Starvation
 - Problem: Math prompts often yield uniform batch outcomes (all correct or all incorrect).
 - Impact: Causes zero gradients in group-relative calculations, wasting compute and stalling convergence.
 - Length Bias & Reward Noise
 - Problem: Standard loss treats all response lengths equally.
 - Impact: Underweights complex long-chain reasoning and introduces noise via harsh truncation penalties.

- **Clip-Higher**: Decoupled Clipping for Enhanced Exploration
 - The Clip-Higher technique modifies the standard PPO clipping mechanism by decoupling the importance sampling ratio bounds. The modified policy gradient objective becomes:

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right]$$

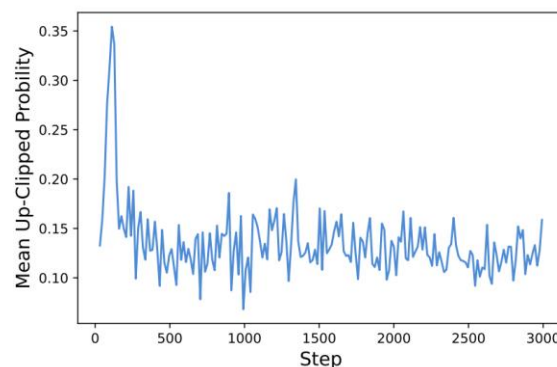
$$\text{s.t. } 0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G.$$



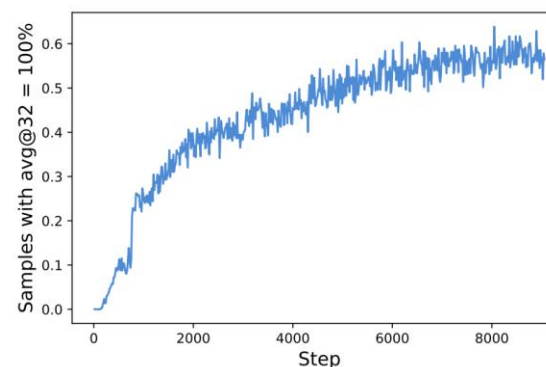
- Dynamic Sampling: Eliminating Zero-Gradient Samples
 - Dynamic sampling addresses gradient starvation by ensuring every prompt in a training batch contributes meaningful gradients. The system over-samples prompts and **filters out** those where all generated responses yield identical rewards (all correct or all incorrect), which would result in zero group-relative advantages.

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right]$$

s.t. $0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G.$



(a) Mean up-clipped probability.



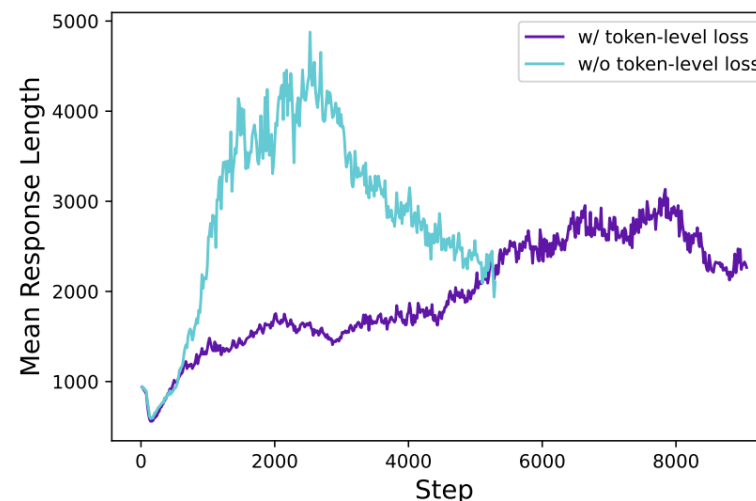
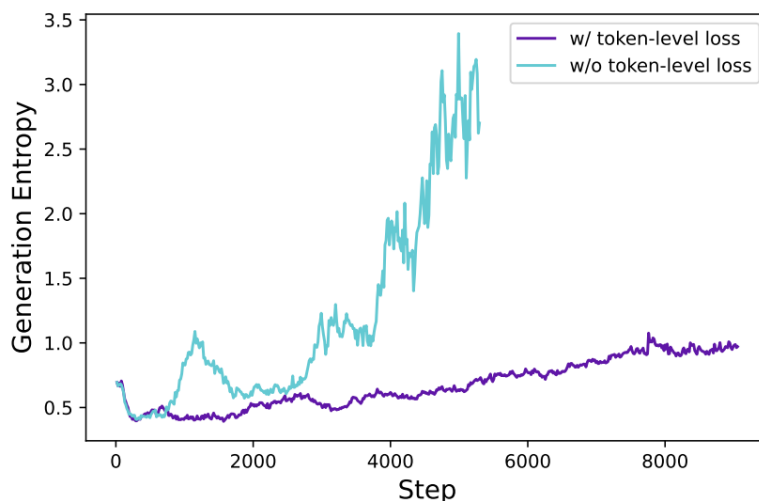
(b) The proportion of samples with an accuracy of 1.

- Token-Level Policy Gradient Loss

- Standard GRPO calculates loss **at the sample level**, averaging token losses within each response before aggregating across samples. This approach gives equal weight to all responses regardless of length, potentially diminishing the influence of longer, more detailed reasoning chains.

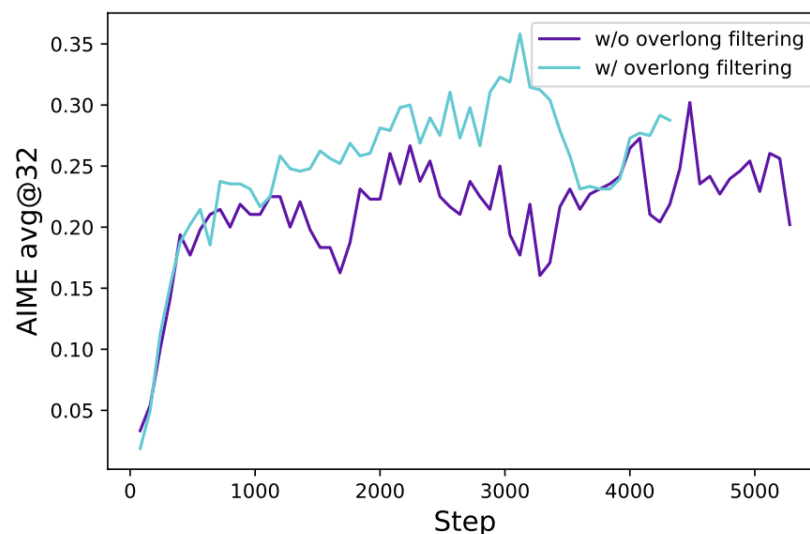
$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right],$$

$$\text{s.t. } 0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G.$$

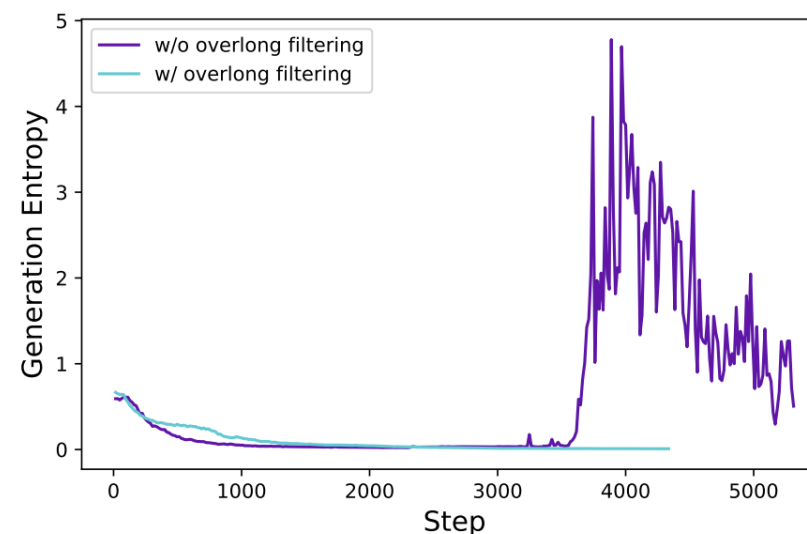


- Overlong Reward Shaping

- To address reward noise from length constraints, DAPO implements soft overlong punishment rather than harsh truncation penalties.
- Responses exceeding a threshold length receive a **graduated penalty** proportional to their excess length, rather than a fixed negative reward.



(a) Performance on AIME.

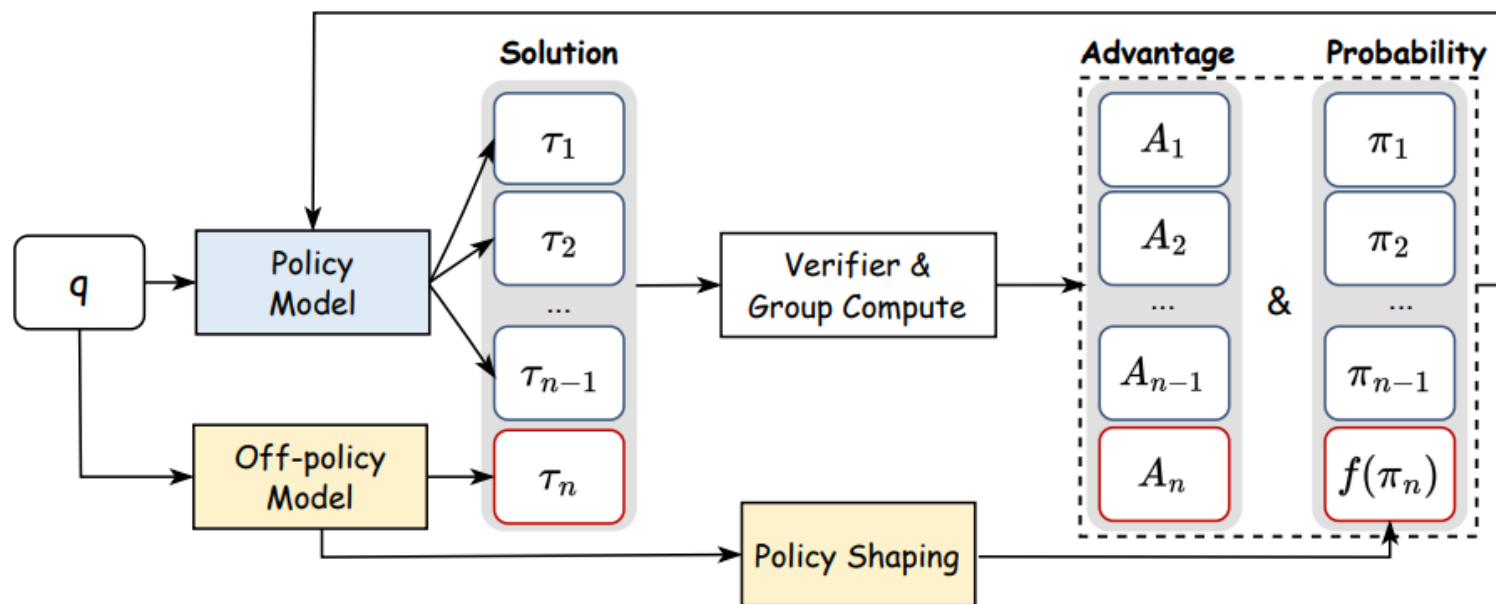


(b) Entropy of actor model.

Off-Policy Optimization



- Current RLVR approaches face a fundamental limitation
 - They operate in an "on-policy" manner, meaning models can only learn from their own generated outputs. This constraint traps models in an "echo chamber" where they can only amplify existing behaviors rather than acquire genuinely novel reasoning capabilities beyond their initial cognitive boundaries.



- On-policy RLVR vs. Supervised Fine-Tuning (SFT)
 - ∅ On-policy Reinforcement Learning
 - ☒ Unified Training & Inference: Unbiased rewards and strong generalization.
 - ☒ Model Constraints: Requires inherent reasoning potential; struggles with sparse rewards otherwise.
 - ☒ Limited Exploration: Difficult to discover rare but high-value reasoning paths; prone to local optima.

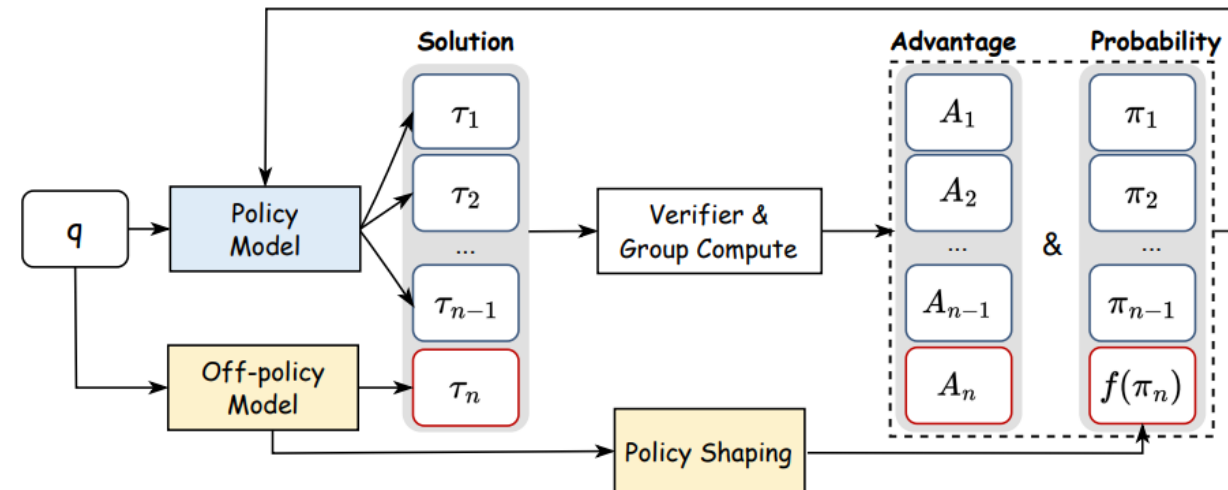


- On-policy RLVR vs. Supervised Fine-Tuning (SFT)
 - ∅ Supervised Fine-Tuning
 -  Efficient Learning: Stable learning from demonstrations with fast convergence and easy implementation.
 -  Lack of Exploration: Unable to derive improvements from new reasoning trajectories.
 -  Poor Generalization: Rote imitation of external demonstrations leads to poor adaptability on unseen task structures.



$$\hat{A}_i = \frac{R(\tau_i) - \text{mean}(\mathcal{G}_{\text{on}} \cup \mathcal{G}_{\text{off}})}{\text{std}(\mathcal{G}_{\text{on}} \cup \mathcal{G}_{\text{off}})},$$

$$\mathcal{J}_{\text{Mixed}}(\theta) = \underbrace{\frac{1}{Z} \left(\sum_{j=1}^{N_{\text{off}}} \sum_{t=1}^{|\tau_j|} \text{CLIP}(\hat{r}_{j,t}(\theta, \phi), \hat{A}_j, \epsilon) \right)}_{\text{off-policy objective}} + \underbrace{\sum_{i=1}^{N_{\text{on}}} \sum_{t=1}^{|\tau_i|} \text{CLIP}(r_{i,t}(\theta), \hat{A}_i, \epsilon)}_{\text{on-policy objective}},$$



Part I: **Off-policy** Data Update – Leveraging external demonstrations to guide learning.

Part II: **On-policy** Data Update – Enabling the model to autonomously explore high-quality strategies.

Advantage computations:

$$\hat{A}_i = \frac{R(\tau_i) - \text{mean}(\mathcal{G}_{\text{on}} \cup \mathcal{G}_{\text{off}})}{\text{std}(\mathcal{G}_{\text{on}} \cup \mathcal{G}_{\text{off}})},$$

Mixed-Policy GRPO:
$$\mathcal{J}_{\text{Mixed}}(\theta) = \underbrace{\frac{1}{Z} \left(\sum_{j=1}^{N_{\text{off}}} \sum_{t=1}^{|\tau_j|} \text{CLIP}(\hat{r}_{j,t}(\theta, \phi), \hat{A}_j, \epsilon) \right)}_{\text{off-policy objective}} + \underbrace{\sum_{i=1}^{N_{\text{on}}} \sum_{t=1}^{|\tau_i|} \text{CLIP}(r_{i,t}(\theta), \hat{A}_i, \epsilon)}_{\text{on-policy objective}},$$

where $\hat{r}_{j,t}(\theta, \phi) = \pi_{\theta}(\tau_{j,t}|q, \tau_{j,<t}) / \pi_{\phi}(\tau_{j,t}|q, \tau_{j,<t})$ and $r_{i,t}(\theta) = \pi_{\theta}(\tau_{i,t}|q, \tau_{i,<t}) / \pi_{\theta_{\text{old}}}(\tau_{i,t}|q, \tau_{i,<t})$.
 $Z = \sum_{j=1}^{N_{\text{off}}} |\tau_j| + \sum_{i=1}^{N_{\text{on}}} |\tau_i|$ is the normalization factor.

Part I: Off-policy Data Update – Leveraging external demonstrations to guide learning.

Part II: On-policy Data Update – Enabling the model to autonomously explore high-quality strategies.

Model	In-Distribution Performance						Out-of-Distribution Performance			
	AIME 24/25	AMC	MATH-500	Minerva	Olympiad	Avg.	ARC-c	GPQA*	MMLU-Pro	Avg.
Qwen-Base [30]	11.5/4.9	31.3	43.6	7.4	15.6	19.0	18.2	11.1	16.9	15.4
Qwen-Instruct [30]	12.5/10.2	48.5	80.4	32.7	41.0	37.6	70.3	24.7	34.1	43.0
Previous RLVR methods										
SimpleRL-Zero [5]	27.0/6.8	54.9	76.0	25.0	34.7	37.4	30.2	23.2	34.5	29.3
OpenReasoner-Zero [7]	16.5/15.0	52.1	82.4	33.1	47.1	41.0	66.2	29.8	58.7	51.6
PRIME-Zero [24]	17.0/12.8	54.0	81.4	39.0	40.3	40.7	73.3	18.2	32.7	41.4
Oat-Zero [6]	33.4 /11.9	61.2	78.0	34.6	43.4	43.7	70.1	23.7	41.7	45.2
Our On-policy RLVR Replication										
On-Policy RL	25.1/15.3	62.0	84.4	39.3	46.8	45.5	82.3	<u>40.4</u>	49.3	57.3
Alternative Off-policy Learning Methods										
SFT	22.2/22.3	52.8	82.6	<u>40.8</u>	43.7	44.1	75.2	24.7	42.7	47.5
RL w/ SFT Loss	19.5/16.4	49.7	80.4	34.9	39.4	40.1	71.2	23.7	43.2	46.0
SFT+RL	25.8/ 23.1	62.7	<u>87.2</u>	39.7	50.4	48.2	72.4	24.2	37.7	44.8
Our Methods										
LUFFY	29.4/ 23.1	<u>65.6</u>	87.6	37.5	57.2	<u>50.1*</u>	80.5	39.9	53.0	<u>57.8*</u>
LUFFY†	<u>30.7/22.5</u>	66.2	86.8	41.2	<u>55.3</u>	50.4*	<u>81.8</u>	49.0	<u>54.7</u>	61.8*

- Qwen2.5-Math-7B:
 - Significant Improvement: Achieved an average score of 50.1 across 6 math benchmarks, surpassing existing RLVR methods by +6.4 points.
 - Strong Generalization: Gained +8.1 points on the newly released AIME 25, validating the transfer and generalization of off-policy reasoning patterns.

Can the model generate multiple answers during testing to improve its reasoning ability?

Self-Refine: Iterative Refinement with Self-Feedback

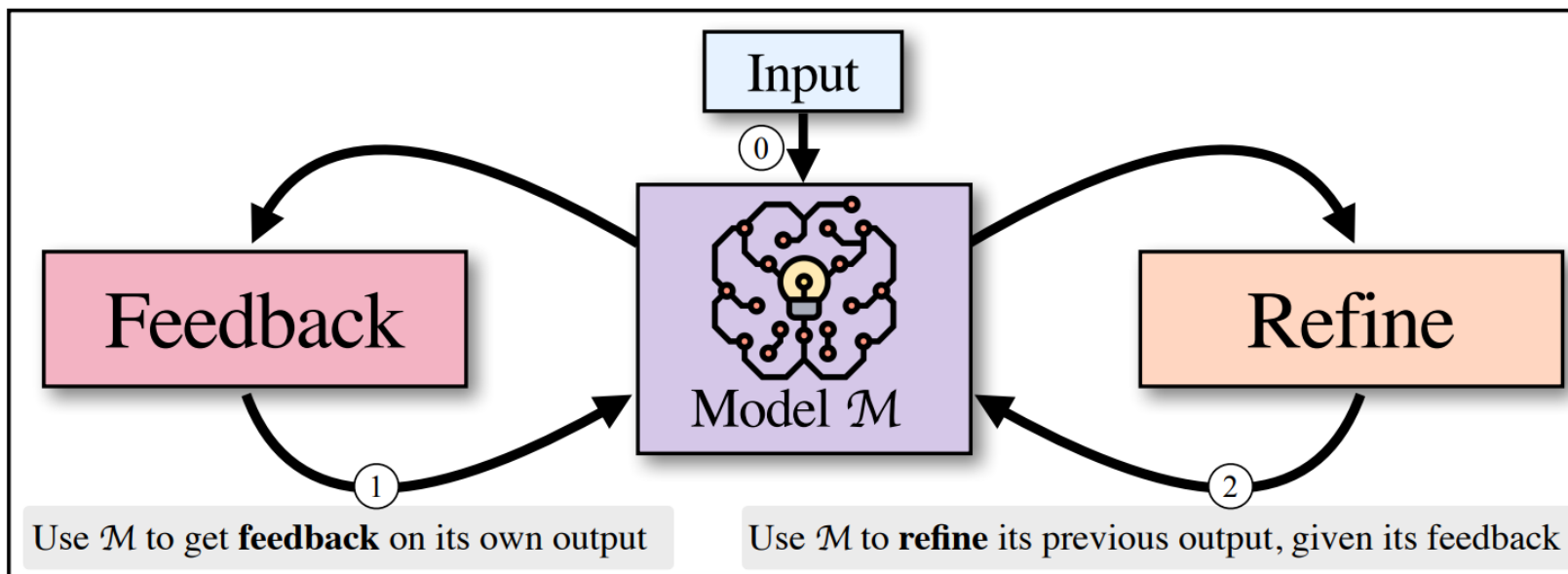


Figure 1: Given an input (①), SELF-REFINE starts by generating an output and passing it back to the same model $\mathcal{M}$ to get feedback (②). The feedback is passed back to $\mathcal{M}$, which refines the previously generated output (③). Steps (②) and (③) iterate until a stopping condition is met. SELF-REFINE is instantiated with a language model such as GPT-3.5 and does not involve human assistance.

- SELF-REFINE introduces an iterative approach that enables large language models to improve their own outputs through a simple feedback loop.
- The method operates entirely within a single LLM using three sequential steps:
 - initial generation,
 - self-feedback,
 - and refinement.
- Unlike traditional approaches that require separate training phases or additional models, SELF-REFINE leverages the few-shot prompting capabilities of modern LLMs to create a self-contained improvement system.

Self-Refine



The core of SELF-REFINE lies in its iterative self-refinement mechanism, which generates specific and actionable feedback that is then used to optimize its own output.

This approach enables the model to learn from past iterations, thus avoiding repeated mistakes and continuously improving with each refinement.

(a) Dialogue: x, y_t

User: I am interested in playing Table tennis.

Response: I'm sure it's a great way to socialize, stay active

(b) FEEDBACK fb

Engaging: Provides no information about table tennis or how to play it.

User understanding: Lacks understanding of user's needs and state of mind.

(c) REFINES y_{t+1}

Response (refined): That's great to hear (...) ! It's a fun sport requiring quick reflexes and good hand-eye coordination. Have you played before, or are you looking to learn?

(d) Code optimization: x, y_t

```
Generate sum of 1, ..., N
def sum(n):
    res = 0
    for i in range(n+1):
        res += i
    return res
```

(e) FEEDBACK fb

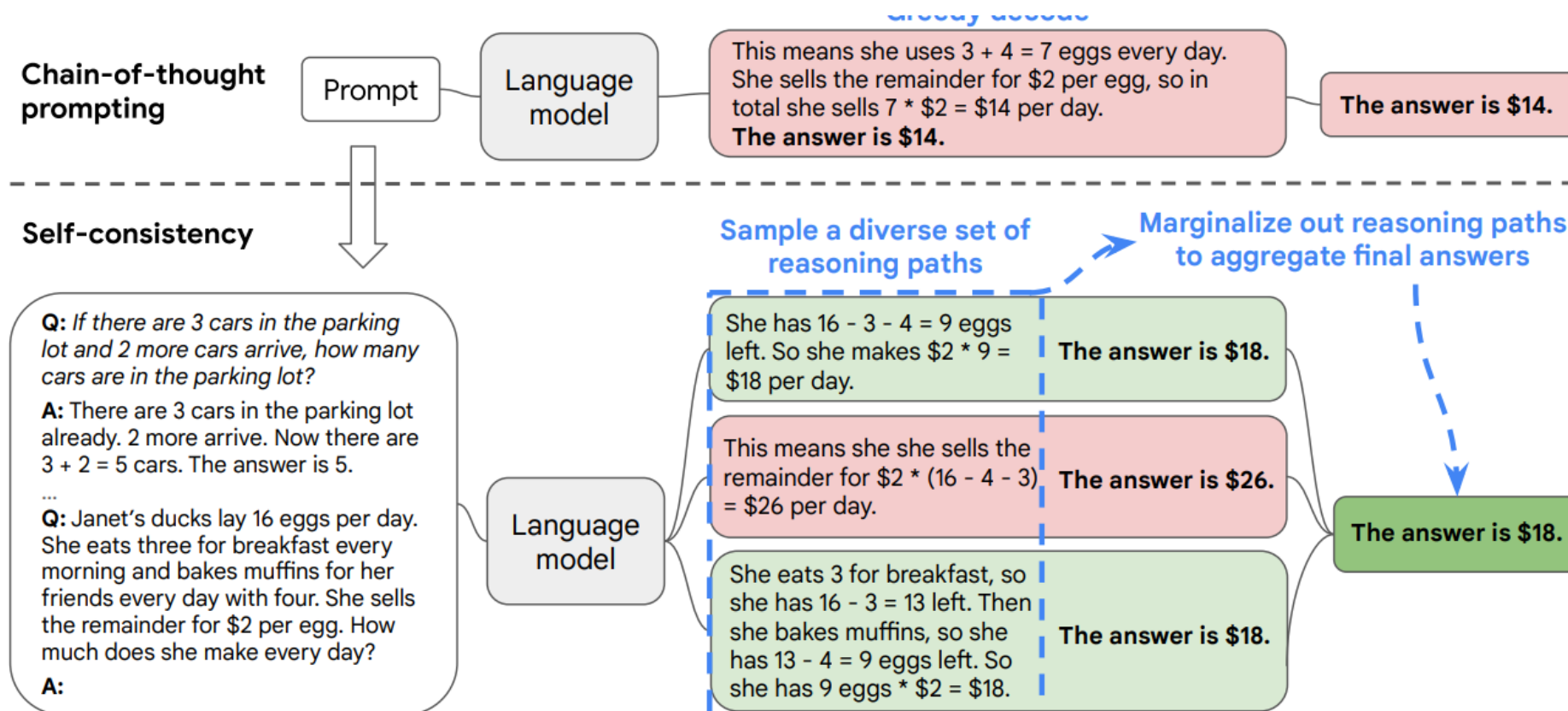
This code is slow as it uses brute force. A better approach is to use the formula ... $(n(n+1))/2$.

(f) REFINES y_{t+1}

```
Code (refined)
def sum_faster(n):
    return (n*(n+1))/2
```

Figure 2: Examples of SELF-REFINE: an initial output generated by the base LLM and then passed back to the same LLM to receive feedback to the same LLM to refine the output. The top row illustrates this for dialog generation where an initial dialogue response can be transformed into a more engaging one that also understands the user by applying feedback. The bottom row illustrates this for code optimization where the code is made more efficient by applying feedback.

Self-Consistency Improves Chain of Thought Reasoning in Language Models

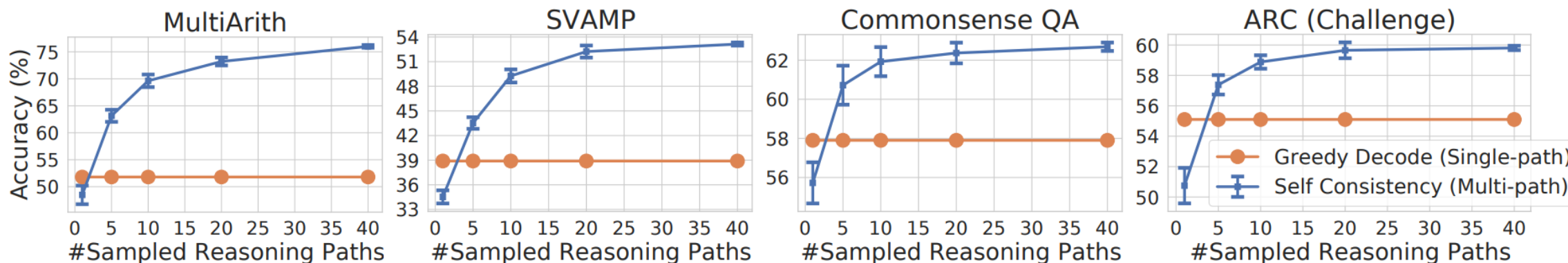


- **Self-Consistency Algorithm.** The self-consistency method operates through three main steps:
 - **Chain-of-Thought Prompting:** The model receives a prompt with few-shot examples demonstrating step-by-step reasoning, following the standard CoT format.
 - **Diverse Path Sampling:** Instead of greedy decoding, the method samples multiple reasoning paths using techniques like:
 - Temperature sampling ($T=0.7$)
 - Top-k sampling ($k=40$)
 - Nucleus sampling (top-p)
 - **Answer Aggregation:** The final answers from all sampled paths are aggregated through majority voting to select the most consistent answer.

Self-Consistency



- Self-consistency (blue) significantly improves accuracy over CoT-prompting with greedy decoding (orange) across arithmetic and commonsense reasoning tasks, over LaMDA-137B. Sampling a higher number of diverse reasoning paths consistently improves reasoning accuracy.



- [\[2512.01374\] Stabilizing Reinforcement Learning with LLMs: Formulation and Practices](#)



2025-12-04

Stabilizing Reinforcement Learning with LLMs: Formulation and Practices

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Thank you